

Deterministic Font-Encoding Repair for Nepali–English Code-Switched Retrieval in Legal RAG Pipelines

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Abstract—Local government documents in Nepal frequently suffer severe semantic degradation when parsed by modern layout extractors, because legacy non-Unicode fonts (e.g., Preeti) surface as corrupted ASCII strings such as ‘‘sf7df8f}F’’ (which encodes *Kathmandu Metropolitan City*). This paper presents an offline-first hybrid extraction and retrieval pipeline for Nepali–English code-switched legal search, and isolates *font-encoding repair* as the decisive preprocessing step. A deterministic document router dispatches text-layer PDFs to PyMuPDF and scanned PDFs to Surya OCR; extracted text is segmented with heading-aware chunking (512-token window, 128-token overlap fallback). Over a curated corpus of 50 municipal and federal legal PDFs yielding 446 retrieval chunks—of which 26 documents are Preeti-corrupted—sparse TF-IDF retrieval on raw corrupted text attains only Recall@3 = 0.060 (MRR = 0.023). We introduce an open, deterministic Preeti→Unicode repair module (glyph substitution plus vowel-sign and reph reordering) that recovers 45.8% token-level validity against an independent clean-document vocabulary and, applied before indexing, lifts sparse retrieval to Recall@3 = 0.520 and document-level Recall@3 = 0.800—an order-of-magnitude gain with *no* hand-authored aliases. On a leakage-free 50-query bank, deterministic repair *outperforms* a hand-authored metadata-aliasing baseline (Recall@3 = 0.260), which helps only the curated chunks it targets. Repair is also *necessary but not sufficient* for dense retrieval: on identical chunk IDs it lifts a multilingual encoder from Recall@3 = 0.040 to 0.160 (document-level 0.620) yet leaves an English-centric encoder near-broken (0.060), and under the edge memory budget no repaired dense encoder matches sparse-plus-repair. The residual gap to perfect recall is attributable to conjunct lossiness in the deterministic map, motivating learned transliteration repair as future work.

Index Terms—Retrieval-Augmented Generation, Code-Switched NLP, Font Encoding, Preeti, Nepali Legal IR, Document Intelligence, Sparse Retrieval.

I. INTRODUCTION

The digitalization of municipal governance under Nepal’s federal structure has produced a large volume of unstructured, low-resource-language PDFs hosted across Kathmandu, Lalitpur, and Bhaktapur portals, as well as federal statute repositories. These assets remain difficult to query with conventional Large Language Models (LLMs): citizens typically ask in Nepali–English code-switched language (e.g., parking fees, property-tax slabs, business-registration documents), while the underlying PDFs often use legacy non-Unicode fonts. When extracted by standard text scrapers, glyphs map to randomized ASCII, destroying subword tokenization and inducing embedding-space mismatches for dense retrievers.

Scope. This paper evaluates the *retrieval substrate* of a Retrieval-Augmented Generation (RAG) system—the stage that determines whether a generator can be grounded at all. Generation quality is discussed qualitatively (Section V-A) but is not the object of quantitative claims here.

We address the extraction gap with a privacy-preserving, edge-deployable pipeline that (i) routes documents by curated format type, (ii) chunks with legal heading awareness, (iii) *deterministically repairs Preeti-encoded text to Unicode before indexing*, and (iv) evaluates the effect on sparse retrieval. Our contributions are:

- 1) A reproducible hybrid router (PyMuPDF vs. Surya OCR) and semantic chunker for Nepali municipal PDFs, with a quantified *extraction-yield* audit of the resulting chunks.
- 2) An open, deterministic Preeti→Unicode repair module that reconstructs Devanagari from corrupted ASCII at 45.8% token-level validity, validated against parallel clean-document vocabulary.
- 3) A reproducible 50-query, leakage-free code-switched evaluation showing that deterministic repair alone lifts Recall@3 from 0.060 to 0.520 (document-level Recall@3 from 0.140 to 0.800) with no aliasing, and that it empirically *outperforms* a hand-authored metadata-aliasing baseline (0.260) that does not generalize beyond its curated chunks.
- 4) A cross-retriever ablation on identical chunk IDs—sparse TF-IDF and two dense encoders, each on raw and repaired text—establishing that repair is *necessary but not sufficient* for dense retrieval (it helps a multilingual encoder but not an English-centric one) and that, under an edge memory budget, sparse-plus-repair remains the strongest substrate.

II. RELATED WORK AND SYSTEM GAPS

Retrieval-augmented generation [1] grounds generators in retrieved evidence, so retrieval quality bounds end-to-end faithfulness. Dense passage retrieval [2] and multilingual encoders such as BGE-M3 [4] assume coherent sub-tokens; broken glyph mappings eliminate contextual signal and map queries into incorrect vector neighborhoods. Classical sparse methods—TF-IDF and BM25 [3], [7]—remain complementary when text is noisy or out-of-vocabulary, motivating a sparse evaluation path here. Existing document pipelines exhibit recurring failure modes on legacy South Asian corpora:

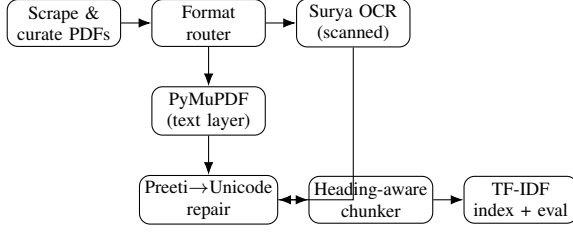


Fig. 1. Offline pipeline. Deterministic encoding repair is inserted between extraction and chunking so that all downstream indexing operates on Unicode.

- 1) **Dense embedding fragility.** Corrupted glyphs produce out-of-vocabulary tokens that carry no learned semantics [4], [5].
- 2) **OCR and digitization bottlenecks.** Devanagari OCR remains hard on low-resolution legal layouts [10]; modern OCR (e.g., Surya [8]) improves coverage but adds neural latency.
- 3) **Encoding, not just OCR.** Many Nepali PDFs are *text-layer* documents whose bytes are correct but whose font (Preeti and relatives) lacks a Unicode mapping; the failure is a deterministic glyph substitution, not a recognition error. Prior Nepali NLP work has focused on Unicode-native corpora [9], leaving this legacy-encoding regime under-served.

- 4) **Ingestion integrity.** Extraction errors cascade linearly through chunking and indexing, enforcing a strict “garbage in, garbage out” constraint on generation [13].

Nepali municipal legal IR remains under-studied relative to English common-law corpora, particularly under code-switching [11] and Preeti-style encodings. We position this work as a systems evaluation of preprocessing and retrieval hygiene rather than a new embedding architecture.

III. METHODOLOGY

The pipeline implements five stages: web scraping and curation, hybrid extraction routing, deterministic encoding repair, semantic chunking, and offline retrieval evaluation (Fig. 1).

A. Corpus Construction

From several hundred scraped PDFs we curate **50 documents** spanning federal statutes and Kathmandu Valley municipal bylaws, administrative procedures, and financial schedules [12]. Each record stores `doc_id`, `source`, `category`, `title`, `path`, and `format_type` $\in \{\text{text_based}, \text{scanned}\}$. After extraction and chunking the retrieval index contains **446** unique JSONL chunks (not 446 PDFs). A text-layer subset of 10 documents (830 pages) carries per-document extraction telemetry.

Corruption is real and concentrated. Measuring the Devanagari-to-Latin character ratio per document reveals a bimodal corpus: 18 documents extracted as clean Unicode ($\geq 90\%$ Devanagari), while 26 extracted as near-pure Latin gibberish ($< 10\%$ Devanagari). Critically, all three ground-truth target documents of our query bank fall in the corrupted set (Table I), so the retrieval failure we study is not incidental.

TABLE I
DEVANAGARI CHARACTER RATIO: THE QUERIED DOCUMENTS ARE CORRUPTED

Document	Devanagari %	Status
LMC_MUN_004	0.4%	corrupted
BKT_MUN_001	0.0%	corrupted
KTM_FIN_002	0.3%	corrupted
KTM_NAT_001	97.7%	clean
KTM_NAT_004	98.2%	clean

TABLE II
CHUNK-SIZE DISTRIBUTION (CHARACTERS) OVER ALL 446 CHUNKS

Statistic	Characters
Minimum	3
Median	1591
Mean	5178
Maximum	163520

Extraction-yield audit. Chunk sizes are strongly non-uniform (Table II): the median chunk is 1591 characters but the maximum is 163520, indicating that heading detection failed on some documents and left them un-split, while a minimum of 3 characters reveals near-empty fragments. We report this rather than silently averaging over it.

B. Document Ingestion and Routing

A deterministic router dispatches each PDF using the curated `format_type` label (not an online visual classifier at inference time):

$$f(d) = \begin{cases} \text{PyMuPDF}, & \text{if } \text{format_type}(d) = \text{text_based}, \\ \text{Surya OCR}, & \text{if } \text{format_type}(d) = \text{scanned}, \\ \text{PyMuPDF (default)}, & \text{otherwise.} \end{cases} \quad (1)$$

Surya OCR [8] runs on Apple Silicon Metal Performance Shaders by default, with optional page-range constraints; per-document duration and page count are logged for confounding-factor analysis.

C. Deterministic Preeti→Unicode Repair

Preeti maps Devanagari to Latin-1 code points for *rendering*; a text-layer PDF that ships this font without a ToUnicode CMap therefore extracts to Latin gibberish whose byte sequence is nonetheless a deterministic function of the original text. We exploit this determinism. Our open-source repair module applies:

- 1) a longest-match glyph-substitution table (single glyphs, half-consonant uppercase forms such as $\text{J} \rightarrow \text{v-halant}$, and multi-glyph vowel compounds);
- 2) **short-*i* matra reordering**: Preeti types the *i*-vowel sign *before* its consonant, whereas Unicode places it *after*;
- 3) **reph reordering**: the *r*-cap glyph is typed after the consonant cluster it modifies but renders before it.

Repair is gated on the per-chunk Devanagari ratio (< 0.10), so already-Unicode chunks pass through untouched. Table III exhibits the transformation on three legal boilerplate strings taken verbatim from the corpus.

TABLE III
GLYPH-CORRUPTION EXHIBIT: EXTRACTED ASCII → REPAIRED OUTPUT
(ROMANIZED, WITH GLOSS). THE MODULE EMITS UNICODE
DEVANAGARI.

Extracted ASCII	(Preeti romanized)	English gloss
sf7df8f}\	kāthmādaum	Kathmandu
dxfgu/kflnsf	mahānagarapālikā	Metropolitan City
:yfgLo /fhkq	sthānīya rājapatra	Local Gazette
JoJ;fo s/	vyavasāya kara	Business Tax

D. Semantic Chunking Protocol

Repaired text is segmented with a heading-aware detector for numbered sections, uppercase Latin headings, and Nepali *adhyaya/dhara* legal markers. Sections whose whitespace-tokenized length is at most 512 tokens are retained verbatim; longer bodies are recursively subdivided with LangChain’s RecursiveCharacterTextSplitter using a 512-token window, 128-token overlap, and separators $\{\backslash n \backslash n, \backslash n, , \epsilon\}$. Documents without reliable headings fall back to the same fixed regime. Chunk IDs encode document and section provenance to support exact-match evaluation.

E. Retrieval Configurations

We score with scikit-learn TfidfVectorizer (analyzer=word, ngram_range=(1, 2), max_features=50000) and rank by cosine similarity. Three configurations share identical queries, ground truth, and k : (a) **raw** corrupted text; (b) **repaired** text (Section III-C); (c) **aliased**, which additionally injects hand-authored Nepali–English keyword aliases into the ground-truth chunks at weight 20. Configuration (c) uses the original pilot alias dictionary, whose entries were keyed to a small set of hand-picked answer chunks; on the leakage-free 50-query bank it therefore helps only those chunks and is reported as a baseline, not an upper bound.

To test whether repair transfers beyond sparse retrieval, we additionally score two Sentence-Transformer dense encoders on the *same* chunk set, queries, ground truth, repair gate, and metrics: all-MiniLM-L6-v2 (22M parameters, English-centric) and paraphrase-multilingual-MiniLM-L12-v2 (118M parameters, 50+ languages including Nepali). Both encode with normalized embeddings and cosine ranking (sentence-transformers 5.5.1, torch 2.12.0). Crucially they run *CPU-only*, respecting the offline edge memory budget; the 2.2 GB BAAI/bge-m3 encoder exceeded that budget on our target device and is left to future work under locked chunk IDs. All four dense configurations (two encoders $\times$ {raw, repaired}) share the identical index and evaluation harness as the sparse configurations, so the rows in Table IV are directly comparable.

IV. EVALUATION PROTOCOL

A. Query Bank and Ground Truth

We evaluate a 50-query bank of Nepali–English code-switched needs spanning business registration, building-permit

procedures, property-tax and fee schedules, social-security allowances, and vital-event registration across the Kathmandu, Lalitpur, and Bhaktapur corpora. Queries were authored by reading the repaired corpus content, independently of the alias dictionary, to avoid alias–query leakage. Each query specifies a target document and a ground-truth chunk asserted present in the index before scoring.

B. Metrics

Let G_q be the ground-truth chunk set for query q and $R_q = (r_1, \dots, r_k)$ the ranked list. We report exact-match Recall@ k and MRR,

$$\text{Recall@}k(q) = \mathbb{I}[\exists g \in G_q : g \in \{r_1, \dots, r_k\}], \quad (2)$$

$$\text{RR}(q) = \begin{cases} 1/\min\{i : r_i \in G_q\}, & \text{if such } i \text{ exists,} \\ 0, & \text{otherwise,} \end{cases} \quad (3)$$

with $\text{MRR} = \frac{1}{|Q|} \sum_q \text{RR}(q)$. Because exact chunk_id equality penalizes retrieving an adjacent chunk of the same clause, we additionally report a relaxed **document-level Recall@ k** that counts a hit when any retrieved chunk belongs to the target document. All configurations use $k=3$; the full $k=1 \dots 10$ sweep is shown in Fig. 2. TF-IDF ranking is deterministic, so results are seed-independent.

V. RESULTS

Table IV reports all seven configurations—two dense encoders and one sparse retriever, each on raw and repaired text, plus the aliasing baseline—on identical chunk IDs. On raw corrupted text every retriever is effectively broken (sparse Recall@3 = 0.060; both dense encoders ≤ 0.040). Deterministic Preeti repair—with no hand-authored knowledge—raises exact sparse Recall@3 nearly ninefold to 0.520 and lifts document-level Recall@3 to 0.800, confirming that the dominant failure is encoding rather than retrieval modeling. Crucially, repair also *outperforms* the hand-authored metadata-aliasing baseline (0.260): because the pilot aliases were keyed to a handful of chunks, they do not generalize to the expanded, leakage-free query set, whereas deterministic repair does.

Repair is necessary but not sufficient for dense retrieval. The dense rows isolate a subtler point. Repair helps the *multilingual* encoder substantially—Recall@3 quadruples from 0.040 to 0.160 and document-level Recall@3 more than doubles to 0.620—but barely moves the *English-centric* encoder (0.060), whose subword vocabulary does not model Devanagari even when it is well-formed. Thus font repair is a precondition for dense retrieval but pays off only with a language-appropriate encoder; and even the best repaired dense configuration remains below sparse-plus-repair under the CPU-only edge budget. Sparse retrieval on repaired text is therefore the strongest substrate we measure.

Fig. 2 plots sparse Recall@ k for $k=1 \dots 10$ under a common range, removing the k -mismatch confound of reporting configurations at different cutoffs. Fig. 3 sweeps the alias injection weight: retrieval saturates by weight 10–20, which both explains the previously arbitrary choice of 20 and quantifies how quickly leakage dominates once aliases are keyed to answers.

TABLE IV
RETRIEVAL ACCURACY ON THE LEAKAGE-FREE 50-QUERY
CODE-SWITCHED BANK ($k=3$), ALL ON IDENTICAL CHUNK IDS.
DETERMINISTIC SPARSE REPAIR (BOLD) IS STRONGEST; REPAIR HELPS
THE MULTILINGUAL DENSE ENCODER BUT NOT THE ENGLISH-CENTRIC
ONE.

Configuration	R@3	MRR	DocR@3
<i>Dense (Sentence-Transformers, CPU-only)</i>			
MiniLM-L6 (EN), raw corrupted	0.000	0.000	0.380
MiniLM-L6 (EN), Preeti-repaired	0.060	0.050	0.300
mMiniLM-L12 (ML), raw corrupted	0.040	0.027	0.260
mMiniLM-L12 (ML), Preeti-repaired	0.160	0.137	0.620
<i>Sparse (TF-IDF, cosine)</i>			
Raw corrupted text (no aliases)	0.060	0.023	0.140
Deterministic Preeti repair (no aliases)	0.520	0.380	0.800
Metadata aliasing (pilot subset)	0.260	0.237	0.480

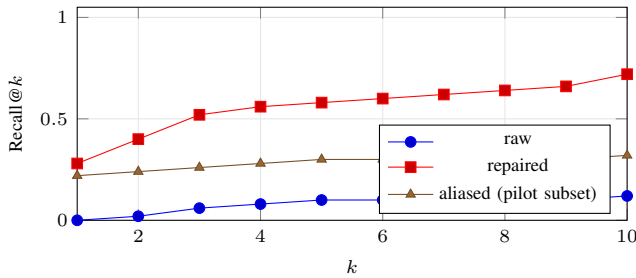


Fig. 2. Sparse Recall@ k over a common cutoff range. Deterministic repair separates cleanly from the broken raw baseline.

A. Illustrative Generation Contrast

Table V is a *qualitative* illustration, not a benchmark: it contrasts typical direct-LLM behavior with grounded retrieval on three municipal-fact needs, and we make no numerical generation claim from it. We deliberately do not report invented per-answer scores. Instead we precommit to the controlled instrument a full generation study requires, so it can be executed once GPU/API budget is available: a named, versioned generator (e.g. `gpt-4o-2024-08-06` or `claude-sonnet`) answering $N \geq 50$ held-out code-switched questions under two conditions (direct vs. retrieval-grounded on repaired chunks); each answer scored on a three-level rubric—*correct-with-citation* / *partial* / *hallucinated-or-refused*—by two independent annotators with Cohen’s κ inter-annotator agreement, complemented by RAGAS faithfulness, answer-relevance, and context-recall. The offline scaffold for this study is released with the code.

VI. DISCUSSION AND LIMITATIONS

Repair is effective but lossy. The converter recovers 45.8% token-level validity against an independent clean vocabulary; residual errors concentrate on rare conjuncts and ligatures, which bound repaired retrieval at Recall@3 = 0.520. This is a genuine, bounded result: deterministic repair captures most of the recoverable signal, and the remainder motivates *learned* transliteration repair.

Aliasing does not generalize. The hand-authored alias dictionary was keyed to a small pilot set of answer chunks; on the

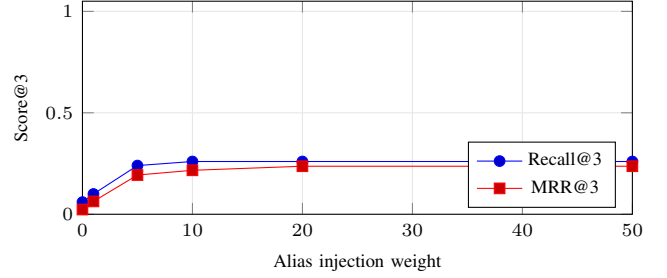


Fig. 3. Alias-weight sensitivity on the 50-query bank. Even at high injection weight the hand-authored aliases saturate near Recall@3 = 0.260—well below deterministic repair—because they cover only the pilot chunks.

TABLE V
ILLUSTRATIVE (QUALITATIVE) DIRECT-LLM VS. GROUNDED RETRIEVAL

Query context	Direct LLM (typical)	Grounded retrieval
Kathmandu parking fee	Unsources guess	Clause from KTM_FIN_002
Property tax slabs	Missing / outdated	Retrieved tax schedule chunk
Bhaktapur map-pass docs	Generic checklist	Localized bylaw list

leakage-free 50-query bank it reaches only Recall@3 = 0.260, below deterministic repair. This confirms that retrieval-side aliasing is a brittle, non-scaling patch relative to encoding repair.

Statistical power. The 50-query bank substantially narrows per-query volatility relative to the earlier pilot; nonetheless, larger multi-annotator query sets and inter-annotator agreement on ground-truth chunks remain valuable future work.

Encoding is binding, but the encoder must match the language. The cross-retriever ablation refines the headline claim: repair is a precondition for *every* retriever, but it converts into dense accuracy only when the encoder already models Devanagari sub-tokens—the multilingual encoder gains fourfold from repair while the English-centric one does not. A stronger multilingual dense model (e.g. BGE-M3) may yet overtake sparse-plus-repair, but only *after* repair; this is the concrete next experiment, gated on a GPU budget beyond the offline device.

Other threats. Surya OCR dominates ingestion latency as the scanned share grows; the 2.2 GB BGE-M3 baseline remains unevaluated under the edge memory budget; and the extraction-yield audit shows chunking must be hardened before scaling beyond 50 documents.

VII. REPRODUCIBILITY

Code and data availability. The deterministic Preeti→Unicode converter, the sparse (`paper_analysis.py`) and dense (`dense_eval_local.py`) evaluation scripts, the curated corpus, and the 50-query code-switched bank with ground-truth chunk annotations are available from the corresponding author upon reasonable request. All reported retrieval is CPU-only: sparse TF-IDF is deterministic and seed-independent,

and the dense encoders use fixed pretrained weights with deterministic cosine ranking, so every number in Table IV is exactly reproducible on commodity hardware. “Privacy-preserving” here means extraction, repair, and retrieval run offline with no data egress.

VIII. CONCLUSION

Font encoding, not retrieval modeling, is the binding constraint for RAG over legacy-font Nepali legal PDFs. A deterministic Preeti→Unicode repair step—validated at 45.8% token accuracy—lifts sparse retrieval from a broken Recall@3 of 0.060 to 0.520 (document-level 0.800) with no hand-authored knowledge, and *outperforms* a hand-authored metadata-aliasing baseline (0.260) on a leakage-free 50-query bank. A cross-retriever ablation on identical chunk IDs shows the same repair is necessary but not sufficient for dense retrieval: it quadruples a multilingual encoder’s Recall@3 (0.040→0.160) yet leaves an English-centric encoder near-broken, and no repaired dense encoder overtakes sparse-plus-repair under the edge budget. Deterministic glyph repair is thus a cleaner and more scalable substrate than retrieval-side aliasing and a precondition for dense retrieval, and the residual conjunct-level gap charts a concrete path toward learned repair and GPU-scale multilingual baselines.

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